Four-Bar EV Charging Mechanism

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***Abstract*—Electric vehicle charging automation requires precise, repeatable mechanical systems. This project presents the design, analysis, and prototyping of an autonomous four-bar mechanism to extend an EV-charging end effector from inside a stationary frame to a vehicle charging port, enabling automated and reliable EV charging. The goal was to produce a mechanically simple, repeatable motion that stays within a 400 mm x 450 mm x 500 mm frame when retracted, exits a slit no larger than 101.6 mm x 101.6 mm, and reaches the charging port center located 152.4 mm from the slit at a height of 240 mm. Kinematic synthesis and constraint optimization using MATLAB determined link geometry and input angles. The mechanism was driven by a single HS-425BB servo and controlled by an Arduino receiving an IR trigger (TSOP38238, command `0x10`). Simulation trajectory error was up to 0.05 mm; the physical prototype (constructed from 6.35 mm MDF) reached the target within approximately 2 mm or the exact intended reach point. IR-trigger reliability was improved with protocol filtering and a 500 ms debounce. A torque vs. input-angle analysis indicates the HS-425BB (spec. $\approx 4.8\ \mathrm{kg\cdot cm} \approx 470\ \mathrm{N\cdot mm}$) supplies sufficient torque for the motion. The system satisfied the project functional requirements and demonstrates robust integration of mechanical synthesis, control, and sensing for autonomous EV charging. MATLAB and CAD simulations indicate that the design would achieve comparable performance if prototyped.**

I. Introduction

A. Motivation and Context

Electric vehicle (EV) charging infrastructure currently relies on manual interaction in many settings. Automating the physical connection to EV charging ports reduces human effort and exposure to hazards, while improving accessibility and repeatability of charging operations. This work demonstrates a compact, reliable mechanical solution that performs a point-to-point reach inside a constrained frame, minimizing software complexity by leveraging a simple four-bar mechanism.

B. Project Objectives and Functional Requirements

The mechanism was designed to meet the following functional specifications:

- Four-bar linkage fits completely inside a 400 mm x 450 mm x 500 mm foam-core frame when retracted.
- End effector must extend through a slit no larger than 101.6 mm x 101.6 mm.
- End effector must reach the charging port center located 152.4 mm from the slit at a height of 240 mm.
- Single HS-425BB servo provides the actuation; Arduino provides control.
- External IR TSOP38238 receiver triggers motion on command `0x10`.

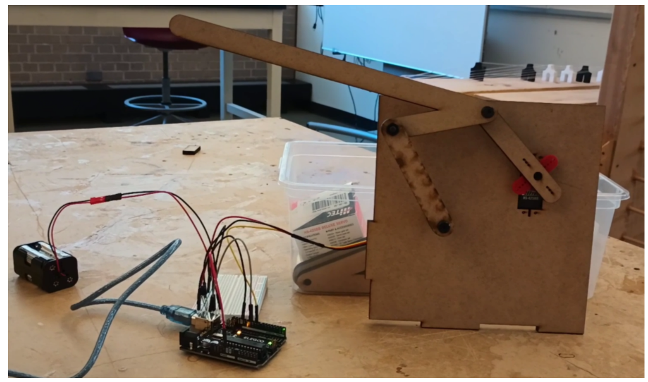


Fig. 1: Prototype used for testing.

- Motion must be smooth and gentle (no deflection of EV), and hold start and end positions for ≥ 3 s.
- Repeatable operation and LED visual indicator during "charging".

Figure 1 shows the prototype evaluated prior to inserting it into the frame for testing.

II. Concept Generation and Evaluation

A. Initial Sketches and Precision Points

The four-bar mechanism geometry was initially defined by assigning arbitrary link lengths in MATLAB. These initial lengths serve as a starting point for MATLAB optimization to minimize end-effector positional error while satisfying frame and slit constraints. The design goal was to move the end effector from a fully retracted position inside the frame to the precision point located 152.4 mm from the slit at a height of 240 mm. The chosen parameterization (inputs to the MATLAB solver) used the following reference geometry (all values in mm):

$$
\begin{aligned}
x_1 &= 280.5 \text{ (pivot x location)},\\
y_1 &= 132 \text{ (pivot y location)},\\
\gamma &= -27.2^\circ, \quad \delta = 25.5^\circ,
\end{aligned}
$$

with initial link lengths:

$$L_{\text{crank}} = 68 \text{ mm}, \qquad L_{\text{coupler}} = 75 \text{ mm}, \tag{1}$$
$$L_{\text{follower}} = 85 \text{ mm}, \qquad L_{\text{ground}} = 87 \text{ mm}, \tag{2}$$

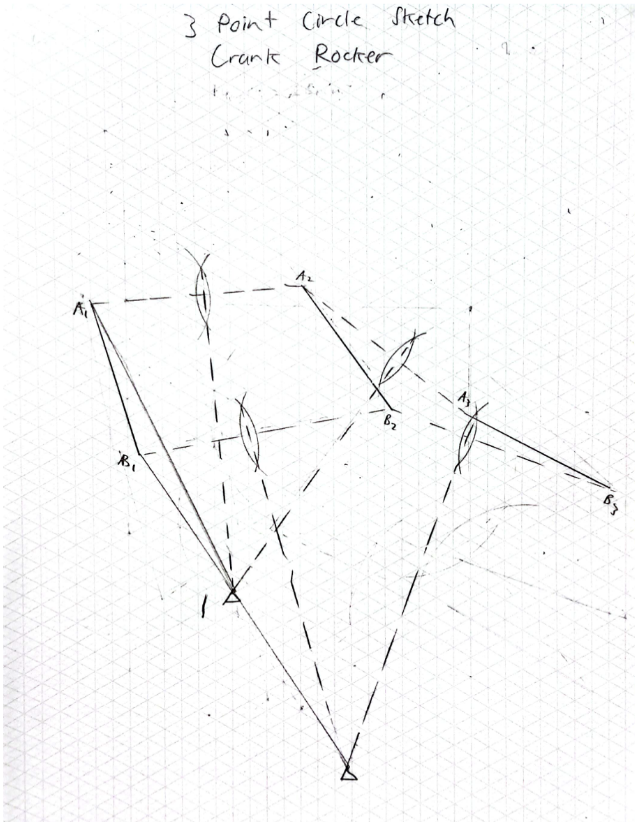


Fig. 2: Three-point circle construction sketches.

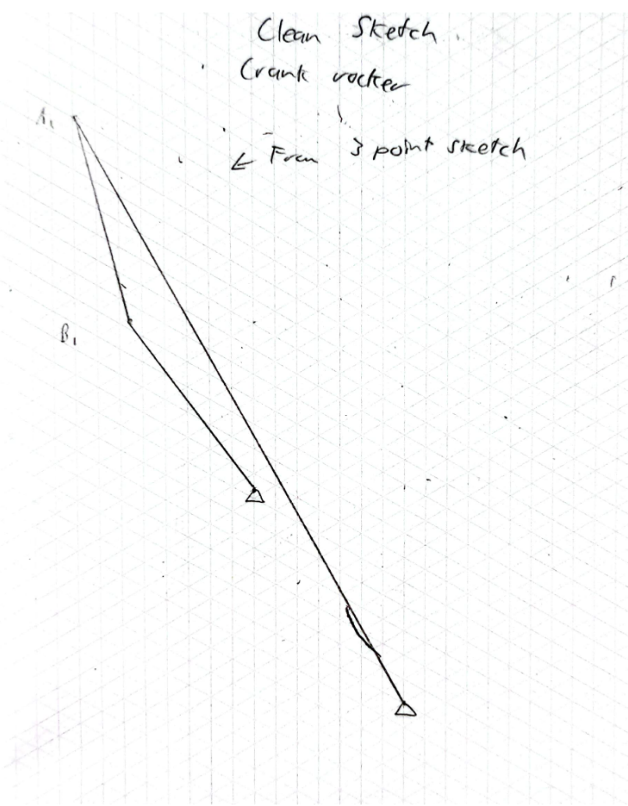


Fig. 3: Clean three-point circle sketch.

TABLE I: Final link geometry used for the prototype (mm).

Component	Length (mm)
Crank (L_2)	68
Coupler (L_3)	75
Follower (L_4)	85
Ground (L_1)	87
End effector reach (r_{ee})	249.3

and end-effector reach $r_{ee} = 249.3$ mm. The start and end servo angles for prototyping were:

$$\theta_{\text{start}} = 157^\circ \quad \text{(retracted)}, \qquad \theta_{\text{end}} = 5^\circ \quad \text{(extended)}.$$

1) Three-Point Circle Sketches: Figure 2 and Figure 3 show hand-drawn three-point circle sketches illustrating the geometric approach.

B. Optimization and Decision Rationale

The MATLAB `fmincon` solver was used with constrained nonlinear minimization to design a four-bar mechanism that meets the functional specifications, including frame clearance, slit passage, and target reach. The objective function minimized the sum of positional errors across the motion path:

$$f(\mathbf{x}) = \sum_{i=1}^{N} \sqrt{(x_{ee,i} - x_t)^2 + (y_{ee,i} - y_t)^2}, \tag{3}$$

where N is the number of discrete crank angles sampled along the motion path, $(x_{ee,i}, y_{ee,i})$ are the end-effector positions computed for each angle $\theta_1^{(i)}$, $\mathbf{x}$ contains the link lengths and pivot coordinates, and (x_t, y_t) is the target position (152.4 mm, 240 mm).

The solver produced link lengths close to the initial guesses, verifying a crank-rocker classification that allows the required crank rotation. The final link geometry used for prototyping is listed in Table I, which shows the lengths directly applied to the MDF prototype. The selection of these values ensured smooth motion, minimal positional error, and compliance with the physical constraints of the frame and slit.

III. Analysis

A. End-Effector Path and Motion

The positions of all links were calculated for each input crank angle θ_1. This produced the path of the end effector from retracted to extended positions. The computed path was compared to the target point $(x_t, y_t) = (152.4 \text{ mm}, 240 \text{ mm})$. The positional difference was ≤ 0.05 mm across the sweep.

Figure 5 shows the initial four-bar geometry before optimization, while Figure 6 demonstrates the optimized trajectory. The optimization refined the link parameters to achieve ≤ 0.05 mm positional error at the target, improving upon the initial design's estimated 5 mm error. The optimization also increased mechanical advantage, decreasing torque.

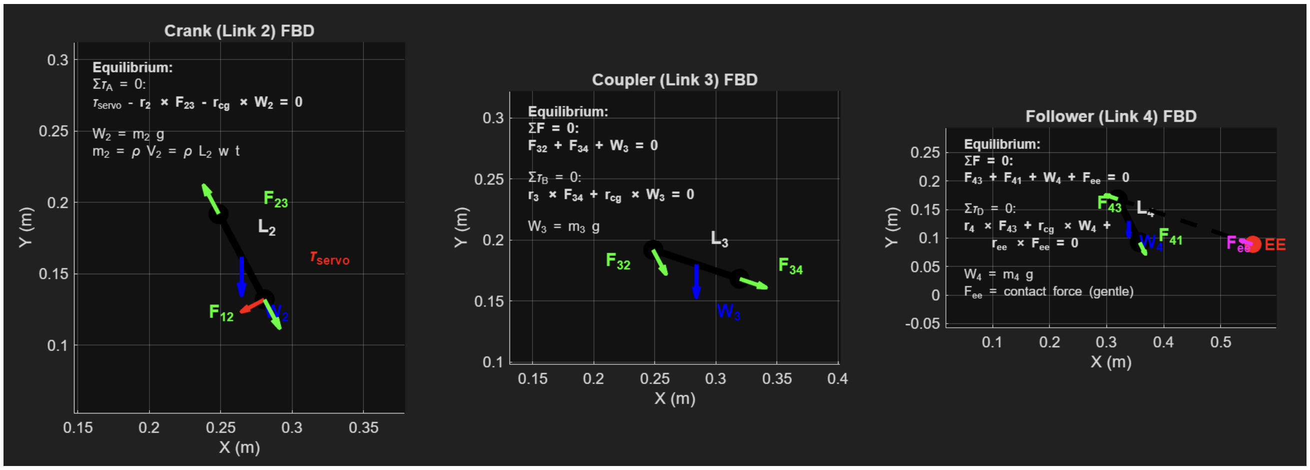


Fig. 4: Free body diagrams showing forces on each link.

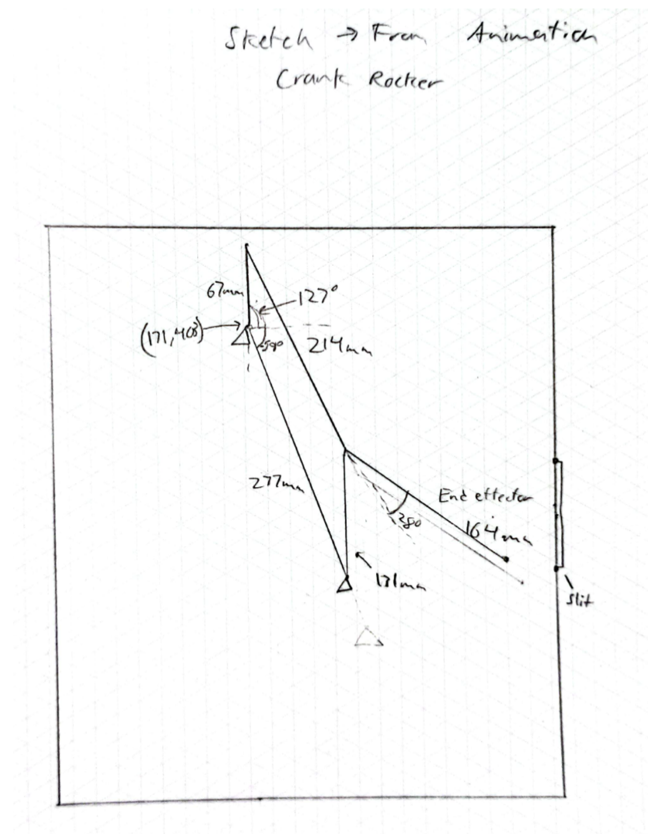


Fig. 5: Four-bar mechanism sketch prior to MATLAB optimization.

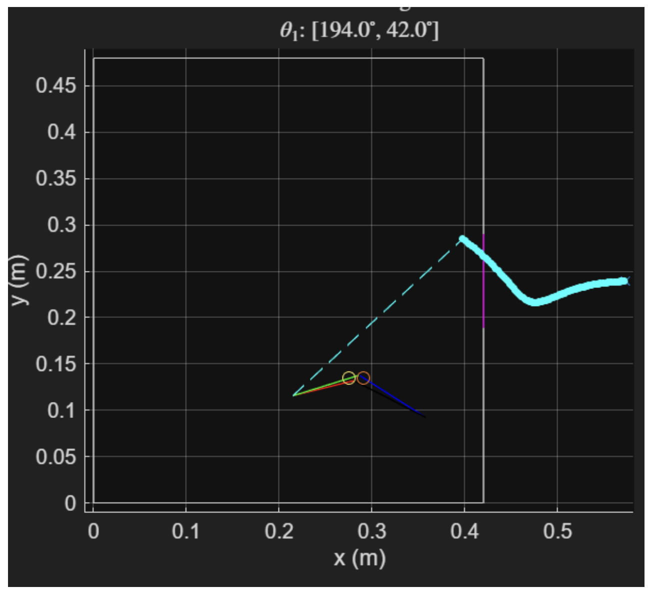


Fig. 6: End-effector trajectory from MATLAB optimization. Target is indicated by cross.

B. Free Body Diagrams

Figure 4 shows the free body diagrams (FBDs) for each link: crank, coupler, and follower. Each diagram illustrates the applied forces, reaction forces, and moments considered in torque analysis. For clarity, all forces are resolved into components along the x and y directions.

- Links approximated as rigid bodies with concentrated masses at centroids (MDF thickness 6.35 mm, typical link widths).
- End-effector contact force assumed small but bounded for worst-case torque.
- Joint friction and tolerances add minor torque margins.

C. Servo Torque Analysis

HS-425BB max torque is $\approx$ 470 N·mm. Figure 8 shows the required servo torque vs input crank angle, with the peak below the servo rating.

D. Trajectory Verification vs Constraints

The end-effector trajectory was checked against functional constraints, including frame boundaries, slit clearance, and target positioning. As shown in Figure 6, the path remains fully within the frame and intersects the target at $(x_t, y_t) =$

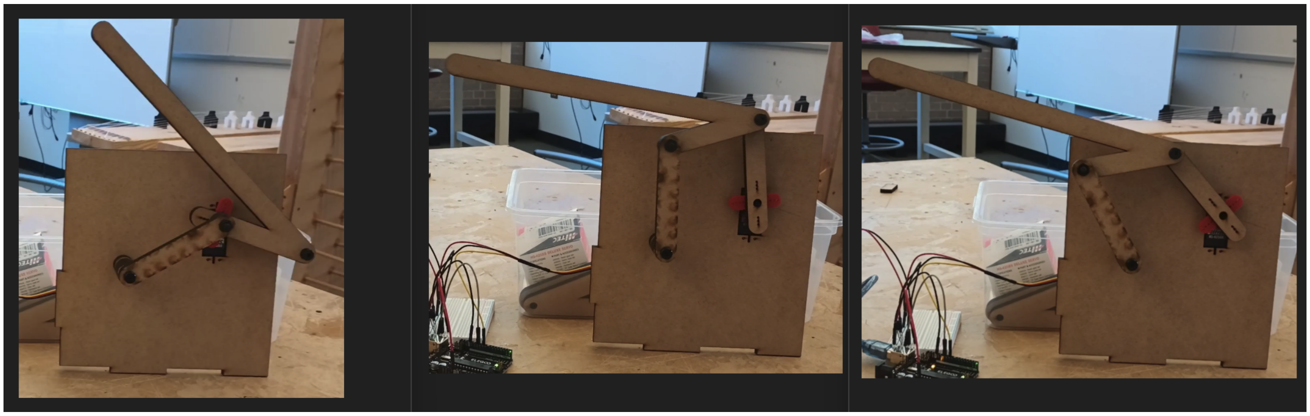


Fig. 7: Experimental sequence: (a) retracted, (b) extending, (c) at charging position.

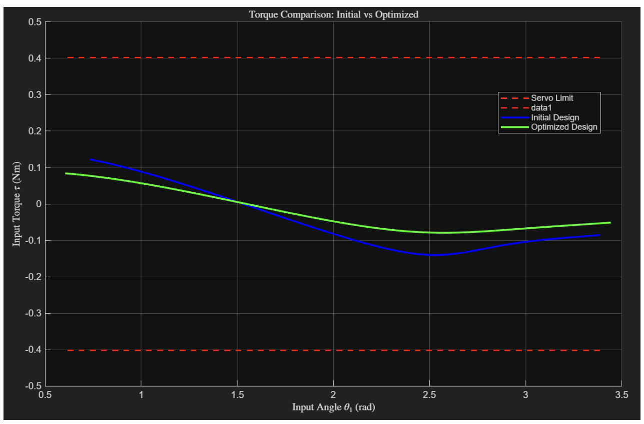


Fig. 8: Required servo torque vs input crank angle.

$(152.4\text{ mm}, 240\text{ mm})$. The mechanism exits the slit correctly without interference. Quantitative evaluation using the constraint functions c and c_{eq} shows a maximum deviation of 0.05 mm, well within the allowed limits, confirming that the design meets all functional requirements.

IV. Experimental Results

The four-bar mechanism prototype was fabricated and tested to evaluate its ability to reach the target position, maintain smooth motion, and comply with frame and slit constraints. Experimental testing involved measuring end-effector position accuracy, servo performance, and reliability over multiple cycles. The results were compared to MATLAB simulations to validate the design.

A. Prototype Fabrication

The prototype links were laser-cut from 6.35 mm MDF and assembled using shoulder screws at the pivots. The completed mechanism was mounted to a foam-core frame for support. Tests indicate that the prototype performance closely matches the MATLAB simulations in terms of end-effector trajectory and positional accuracy (see Fig. 7).

B. Control, Sensing, Reliability

An Arduino Uno controlled the servo and processed the IR input from a TSOP38238 receiver. To ensure reliable signal detection, a 500 ms debounce was implemented in the code to prevent multiple triggers from a single IR pulse. Protocol validation was applied to verify that the received signal matched the expected command pattern before moving the mechanism. This combination of hardware and software filtering ensured robust and repeatable actuation of the end effector.

C. Performance Results

The mechanism successfully reached the target position with an accuracy of ± 2 mm. MATLAB simulations had predicted a positional error of ≤ 0.05 mm, showing that the prototype performance was reasonably close given material compliance. The mechanism completed 5 consecutive actuation cycles reliably without servo stalls. The servo motion was smooth, and the end effector made gentle contact with the target, avoiding excessive impact.

D. Analysis of What Went Right/Wrong

Successes: The mechanism exhibited smooth motion due to 1° stepping increments, which minimized abrupt movements. IR signal filtering and debounce prevented false triggers, ensuring reliable actuation. All link motions remained within the frame constraints throughout testing, and the servo detach strategy effectively eliminated idle noise when the mechanism was stationary.

Issues and Corrections: Initial servo jumps at the start of motion were corrected using the `moveTo()` function in the Arduino code. Spurious IR triggers were mitigated with additional protocol validation and the 500 ms debounce. The observed 2 mm end-effector error was primarily due to MDF compliance and assembly clearances; using stiffer material or tighter tolerances would further reduce this discrepancy.

V. Discussion

A. Verification Against Functional Requirements

Table II summarizes the verification of the four-bar mechanism against its functional requirements. Each requirement was evaluated through experimental testing, including frame clearance, slit exit, end-effector target reach, IR trigger reliability, motion smoothness, hold times, repeatability, and gentle contact with the target. A checkmark (✓) indicates that the prototype satisfied the requirement within the defined tolerance or operating constraints. This table provides a concise overview of the mechanism's compliance with the design specifications.

TABLE II: Functional requirement verification checklist

Requirement	Notes	Pass
Frame retraction	Mechanism fully inside frame	✓
Slit constraint	Motion exits via ≤ 101.6 mm slit	✓
Target reach	152.4 mm at 240 mm height	✓(± 2 mm)
IR trigger	TSOP38238 detects reliably	✓
Smooth motion	moveTo() 1° / 15 ms	✓
Hold times	≥ 3 s	✓
Repeatability	10+ cycles	✓
Gentle contact	No deflection	✓

B. Arduino Control Summary

The mechanism is controlled by an Arduino Uno, which interfaces with the TSOP38238 IR receiver to detect input signals from a remote or IR emitter. Debounce of 500 ms is implemented in the Arduino code to prevent multiple triggers from a single IR pulse. Protocol validation is performed in software to ensure only valid commands actuate the mechanism.

Motion is driven by an HS-425BB servo, which is commanded using the `moveTo()` function from the Servo library. The servo moves in 1° increments every 15 ms to provide smooth and controlled motion, minimizing abrupt movement. LED indicators provide visual feedback for successful command reception and motion execution.

The circuit consists of the Arduino connected to the IR receiver, with the servo powered from a stable 5 V source. Appropriate current-limiting and decoupling capacitors ensure reliable servo operation and prevent voltage dips. This setup provides a compact, reliable control system that accurately reproduces the motions predicted by MATLAB simulations.

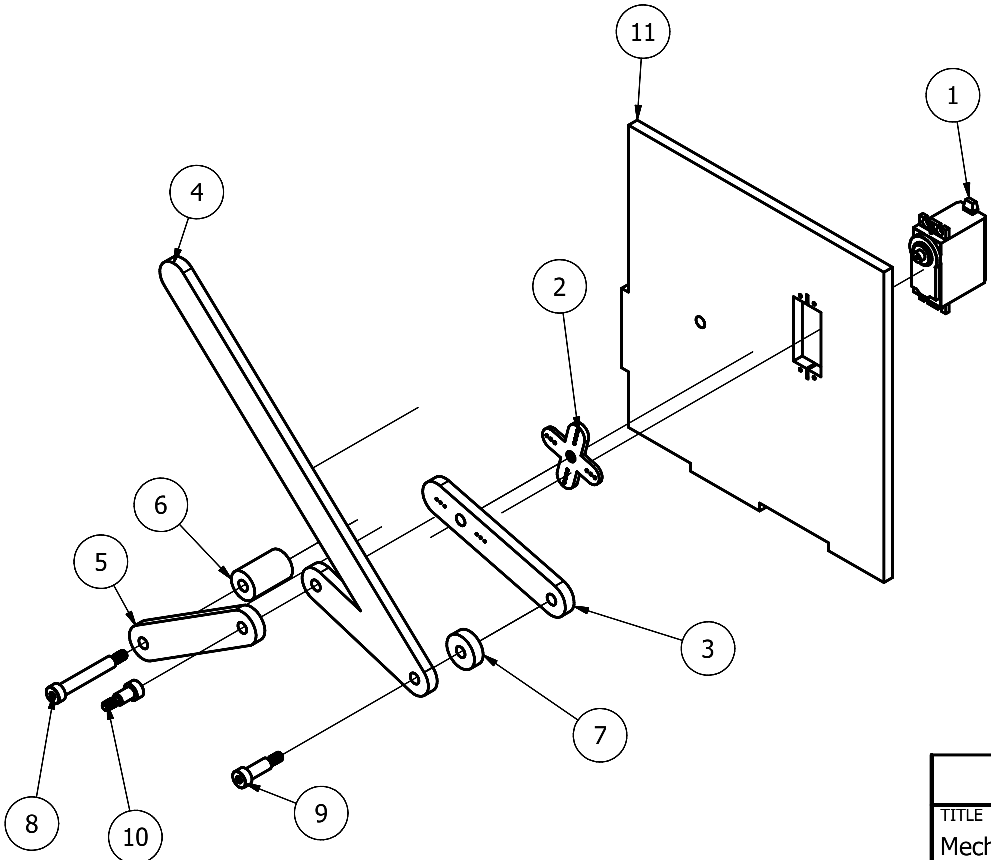
VI. Conclusion

This project presented the design, optimization, and prototyping of a compact four-bar mechanism for autonomous EV charging. Starting from initial sketches and precision point analysis, link lengths and pivot locations were selected and refined using MATLAB `fmincon` optimization to minimize end-effector positional error. Analytical evaluation, including free body diagrams and torque calculations, verified that the HS-425BB servo could drive the mechanism throughout the required motion range while satisfying mechanical and frame constraints.

A physical prototype was fabricated from laser-cut MDF and mounted on a foam-core frame. The mechanism was controlled by an Arduino Uno, which received IR signals from a TSOP38238 receiver, implemented software debounce and protocol validation, and commanded the servo using smooth incremental motion. Experimental testing demonstrated that the end effector reliably reached the target within 2 mm, remained within the frame, exited the slit without interference, and operated repeatably over multiple cycles. Functional verification summarized in Table II confirms that all design requirements—including frame retraction, slit clearance, target reach, smooth motion, and IR-trigger reliability—were satisfied.

Overall, the project demonstrates the effectiveness of integrating kinematic synthesis, numerical optimization, mechanical analysis, prototyping, and control to produce a practical, low-complexity mechanism capable of autonomous EV charging.

Appendix A
Engineering Drawings



PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	Servo Skiba
2	1	Servo Horn 3 Skiba
3	1	Crank Skiba
4	1	Coupler Skiba
5	1	Follower-Skiba
6	1	spacer Skiba
7	1	spacer 2 Skiba
8	1	93897A306_Nylon Shoulder Screws
9	1	93897A272_Nylon Shoulder Screws
10	1	93897A255_Nylon Shoulder Screws
11	1	Cut Pegboard Panel Skiba

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TITLE
Mechanism Drawings

SCALE 1 / 3	DATE 10/14/2025	CLASS MMAE 232
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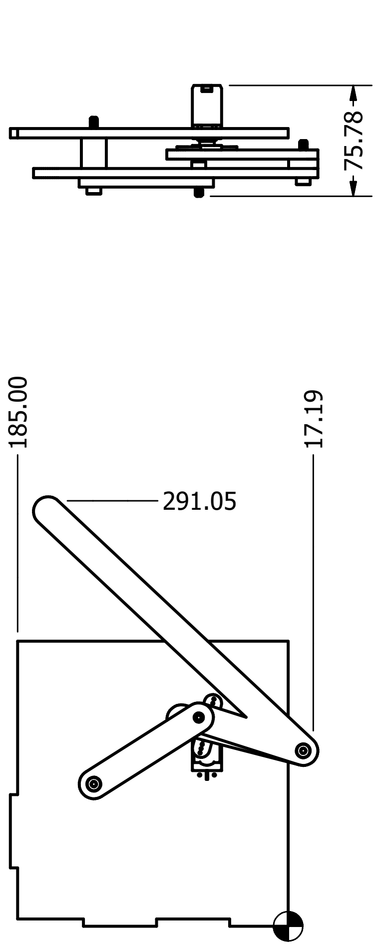

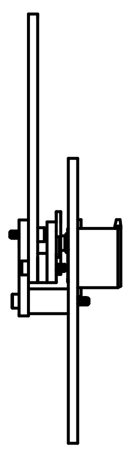

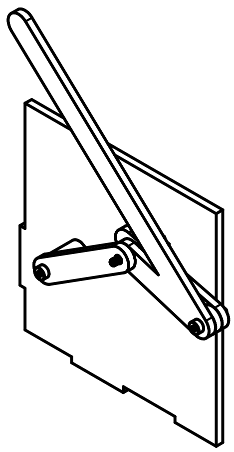

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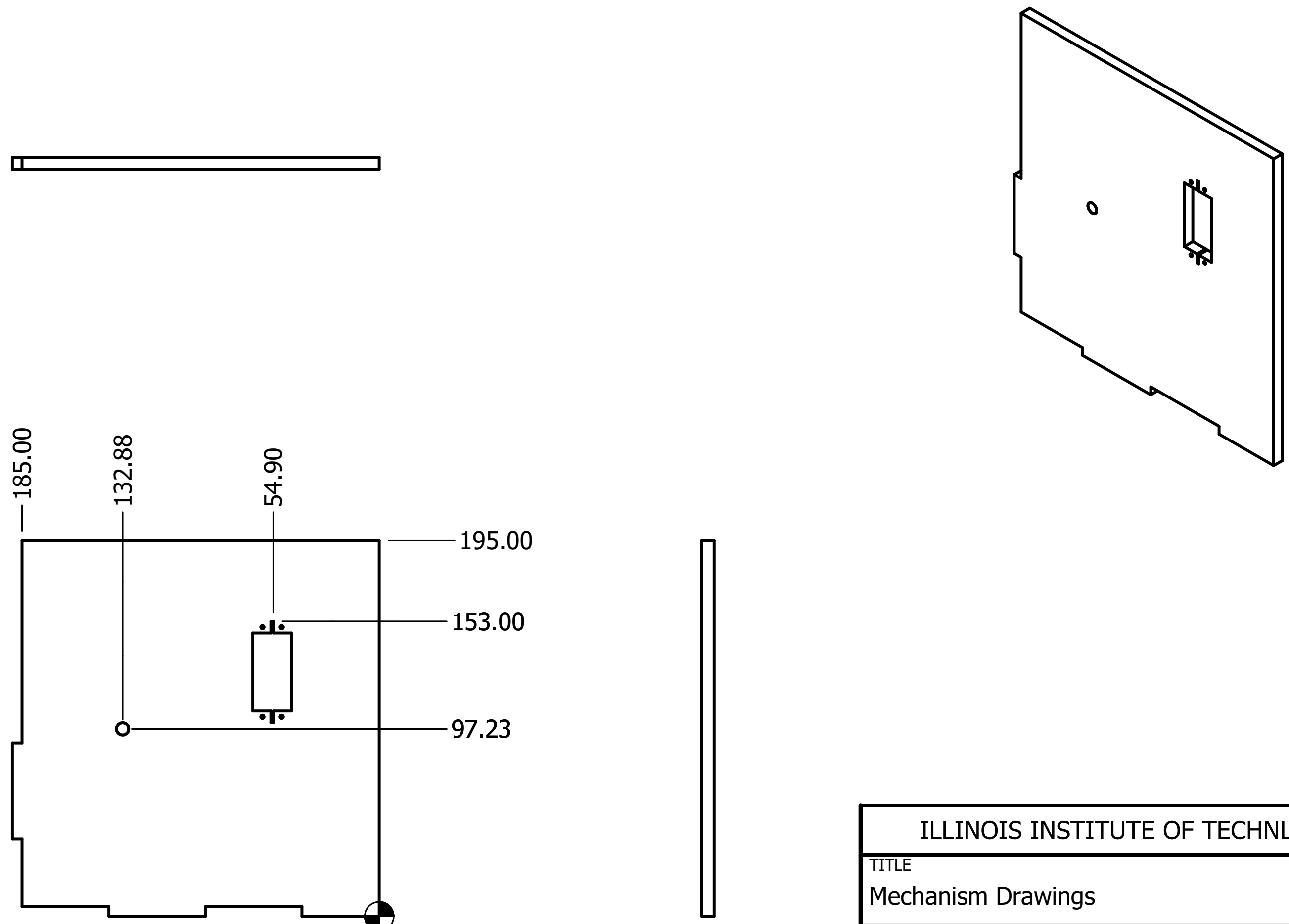
TITLE Mechanism Drawings			
SCALE 1 / 4	DATE 10/14/2025	CLASS MMAE 232	
UNITS mm	DWG NO Cut Four Bar Assembly Without	REV Frame Skib	
DRAWN BY: NICHOLAS SKIBA	SHEET 2 OF 8		



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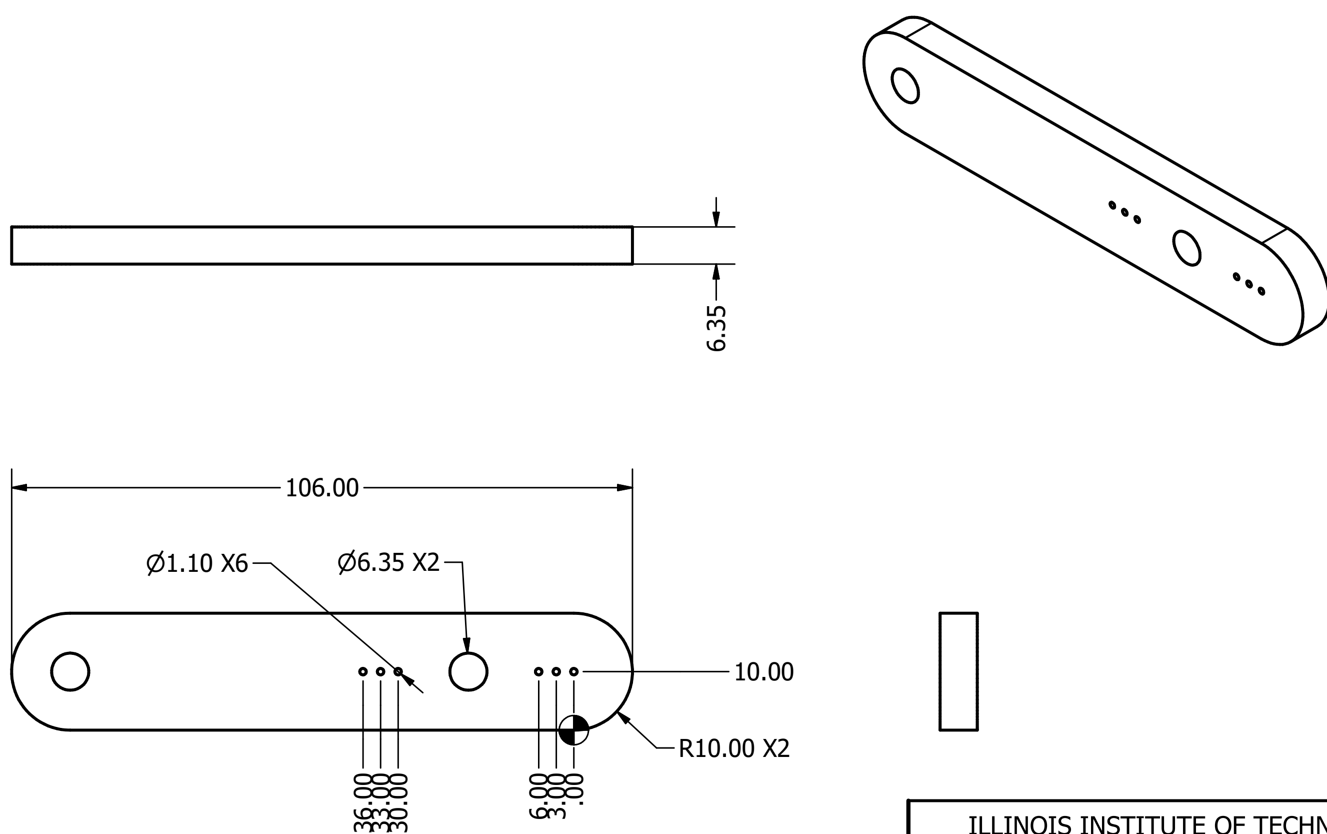

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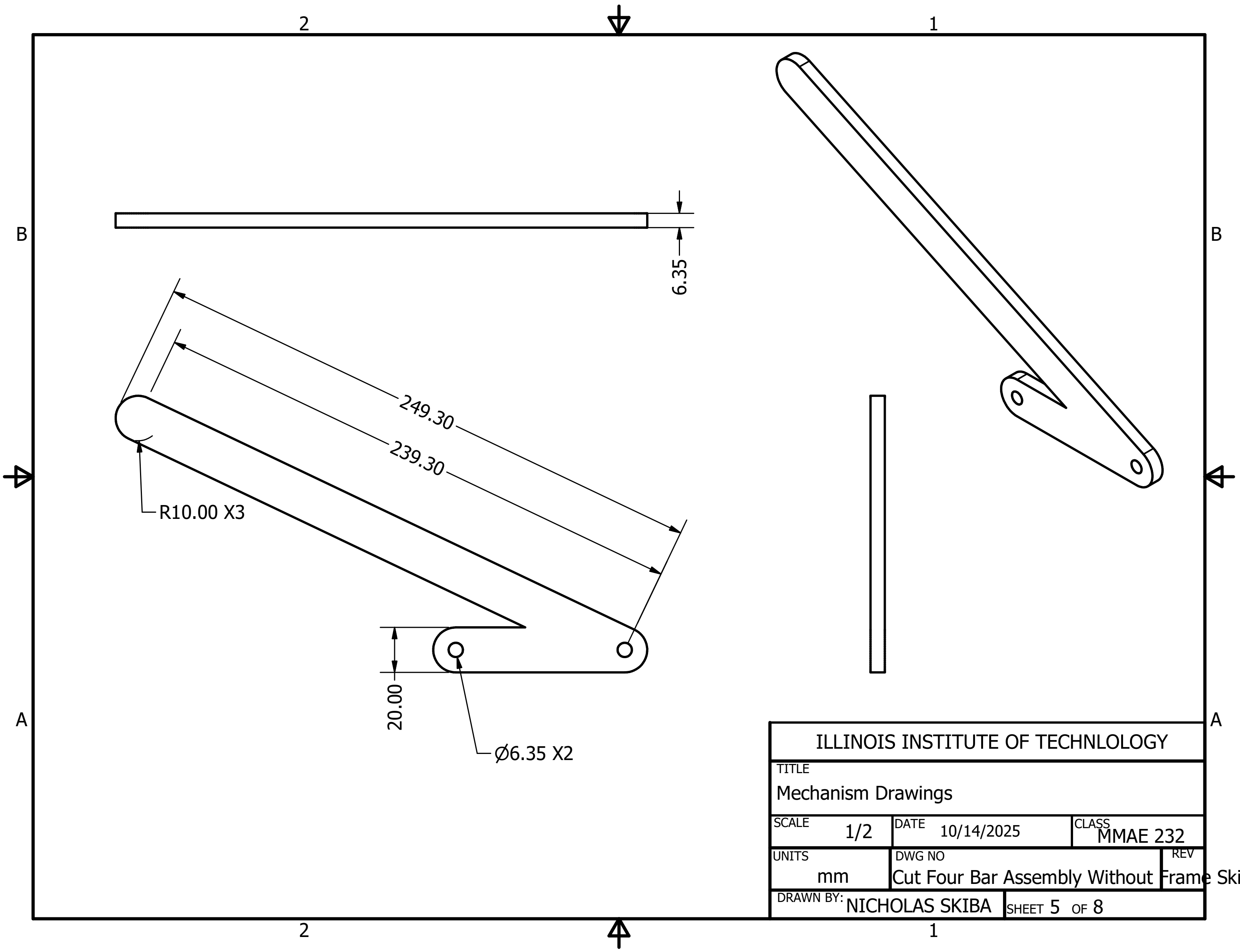

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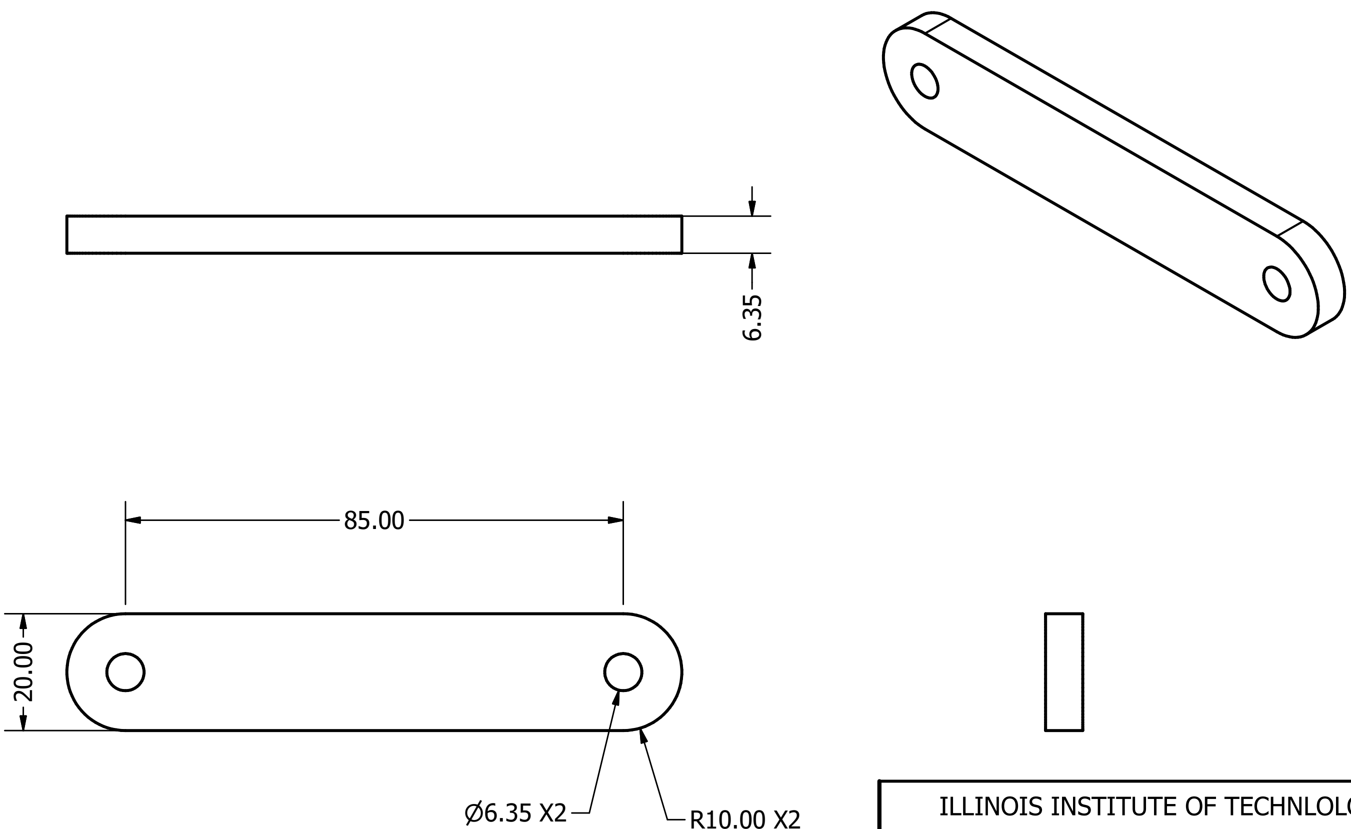

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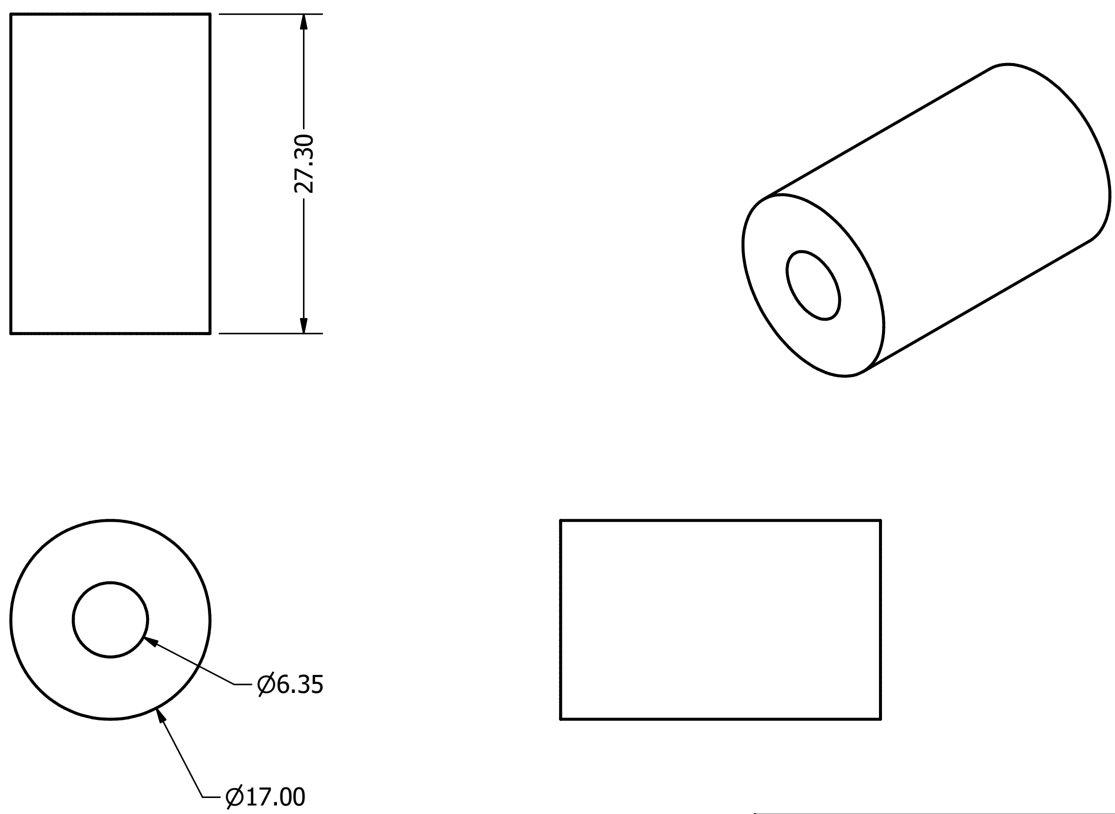

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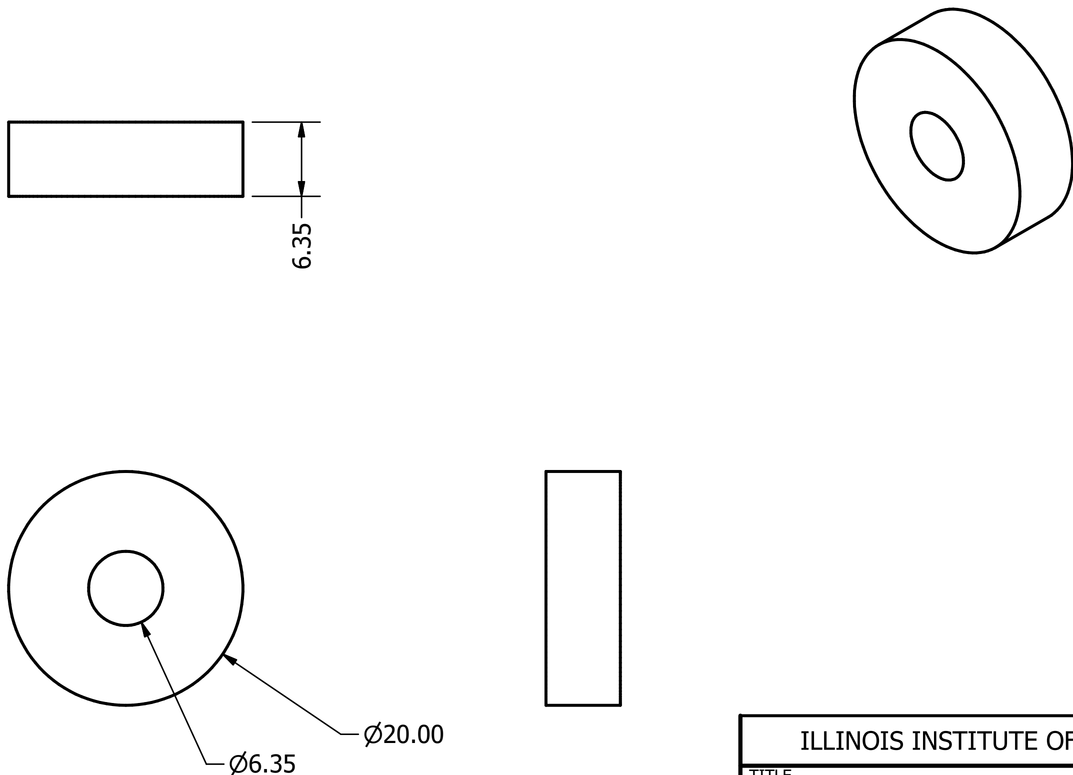

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